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Math 3108 – Linear Algebra
Exam 1 – Part B
Prof. Runnion
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Directions:

- 1) This packet is a continuation of Part A. All directions on Part A still apply.
- 2) This packet has three sheets of paper, including this cover page. Do NOT remove the staple or remove any sheet from this packet.

	4	5	Part B
Points Earned	24	18	42
Maximum Points	24	20	44

**DO NOT OPEN THIS EXAM
UNTIL TOLD TO DO SO**

Name: _____

Multiple Choice and True/False. Circle the correct answer.

9. (2 points) How many pivot columns must a 7×3 matrix have if its columns are linearly independent? $\Rightarrow$ Pivot in Every Column! And Matrix Has 3 Columns

(Circle one) 1 2 3 4 5 6 7

10. (2 points) If $\mathbf{v}_1$ and $\mathbf{v}_2$ are nonzero vectors in $\mathbb{R}^4$ and $\mathbf{v}_2$ is not a scalar multiple of $\mathbf{v}_1$, then $\{\mathbf{v}_1, \mathbf{v}_2\}$ is linearly independent.

(a) Always true (b) Sometimes true (c) Never true

11. (2 points) If the set $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$ is a set of linearly independent vectors in $\mathbb{R}^7$, then the set $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is also linearly independent. $\Rightarrow$ No free Variables $\Rightarrow$ Pivot every Column

(a) Always true (b) Sometimes true (c) Never true

Open Answer. Show ALL your work!

12. (10 points) Find the standard matrix of the horizontal shear transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$

which leaves $\mathbf{e}_1$ unchanged and maps $\mathbf{e}_2$ into $\mathbf{e}_2 + 3\mathbf{e}_1$. Then, find $T\left(\begin{bmatrix} -2 \\ 3 \end{bmatrix}\right)$.

$$\begin{aligned} T(\bar{\mathbf{e}}_1) &\Rightarrow \bar{\mathbf{e}}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\ T(\bar{\mathbf{e}}_2) &\Rightarrow \bar{\mathbf{e}}_2 + 3\bar{\mathbf{e}}_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} + 3\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0+3 \\ 1+0 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \end{bmatrix} \end{aligned} \left. \vphantom{\begin{aligned} T(\bar{\mathbf{e}}_1) \\ T(\bar{\mathbf{e}}_2) \end{aligned}} \right\} \text{Standard Matrix } A = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$$

$$T\left(\begin{bmatrix} -2 \\ 3 \end{bmatrix}\right) = A\bar{\mathbf{x}} = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} -2 \\ 3 \end{bmatrix} = \begin{bmatrix} -2(1) + 3(3) \\ -2(0) + 3(1) \end{bmatrix} = \begin{bmatrix} 7 \\ 3 \end{bmatrix}$$

13. (8 points) Compute the product AB where $A = \begin{bmatrix} 5 & -10 \\ -1 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & -4 \\ 1 & -2 \end{bmatrix}$

$$AB = (2 \times 2)(2 \times 2) = 2 \times 2$$

$$\begin{bmatrix} 5 & -10 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 2 & -4 \\ 1 & -2 \end{bmatrix} = \begin{bmatrix} 10 - 10 & -20 + 20 \\ -2 + 2 & 4 - 4 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Name: _____

14. Consider the system of linear equations

$$x_1 + 3x_2 = 2$$

$$4x_1 + hx_2 = 8$$

- a) (4 points) Find the value(s) of h , if any, such that the system has exactly one solution.

$$\left[\begin{array}{cc|c} 1 & 3 & 2 \\ 4 & h & 8 \end{array} \right] \xrightarrow{-4R_1 + R_2 \rightarrow R_2} \left[\begin{array}{cc|c} 1 & 3 & 2 \\ 0 & -12+h & 0 \end{array} \right] \quad |h \neq 12|$$

- b) (4 points) Find the value(s) of h , if any, such that the system has infinitely many solutions.

$$|h = 12| ; \text{ This would cause there to exist a free variable}$$

$$0x_2 = 0 ; x_2 \text{ is free}$$

15. (6 points) The following *augmented* matrix in echelon form represents a system of linear equations. If the system is consistent, find the general solution in parametric vector form. Otherwise, state that the system is inconsistent and indicate why you came to that conclusion.

$$\left[\begin{array}{ccc|c} (1) & 0 & 4 & 6 \\ 0 & (1) & 4 & 4 \\ 0 & 0 & (1) & 0 \end{array} \right] \xrightarrow{-4R_3 + R_2 \rightarrow R_2} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 6 \\ 0 & 1 & 0 & 4 \\ 0 & 0 & 1 & 0 \end{array} \right] \begin{array}{l} x_1 = 6 \\ x_2 = 4 \\ x_3 = 0 \end{array}$$

Consistent / Inconsistent?

$$\vec{x} = \begin{bmatrix} 6 \\ 4 \\ 0 \end{bmatrix}$$

16. (6 points) Explain, in one or more complete English sentences, why a set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ in $\mathbb{R}^3$ must be linearly independent when $\{\mathbf{v}_1, \mathbf{v}_2\}$ is a linearly independent set and $\mathbf{v}_3$ is not in $\text{Span}\{\mathbf{v}_1, \mathbf{v}_2\}$.

For a set of vectors to be linearly dependent at least one vector has to be a linear combination of the preceding vectors. Because we know $\{\bar{\mathbf{v}}_1, \bar{\mathbf{v}}_2\}$ are independent we know that they are not constant multiples of one another. Furthermore, since we know $\mathbf{v}_3$ is not in the span $\{\bar{\mathbf{v}}_1, \bar{\mathbf{v}}_2\}$. We know that it is NOT a linear combination of the preceding vectors by definition of span. Thus, we know $\{\bar{\mathbf{v}}_1, \bar{\mathbf{v}}_2, \bar{\mathbf{v}}_3\}$ in $\mathbb{R}^3$ must be linearly independent.